The iid Berry–Esseen constant

A computer-assisted proof with exact certificates

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Abstract

We prove that the optimal iid Berry–Esseen constant equals Esseen's constant $C_{\mathrm{E}} = (3+\sqrt{10})/(6\sqrt{2\pi})$ and that $C_n \le C_{\mathrm{E}}$ for every positive integer $n \ge 1$, combining ordinary analytical reductions with finite exact certificates verified end-to-end (`full_pass`). The proof selects an unrestricted extremal law, preserves that law through contact, resource and conditional-count estimates, and covers all positive integers by thirteen compatible contributions. The infinite tail has its own three-loss base and least-failure argument. We give the classical lower approximation separately, prove the binomial normalization extension across one half, and provide a companion source supplement with the longer analytical derivations and their logical dependencies.

1 Scope and target

Let $\mathcal{P}$ be the real probability laws with mean zero, variance one, and finite third absolute moment $\rho(P) = \mathbb{E}_P|X|^3$. For iid variables with law P, let $S_n = X_1 + \cdots + X_n$ and

$$\Delta_n(P) = \sup_{H\in\mathbb{R}} |\mathbb{P}_P(S_n \le H) - \Phi(H/\sqrt{n})|, \qquad R_n(P) = \frac{\sqrt{n}\,\Delta_n(P)}{\rho(P)}.$$

Set

$$C_n = \sup_{P\in\mathcal{P}} R_n(P), \qquad C^* = \sup_{n\ge 1} C_n, \qquad C_{\mathrm{E}} = \frac{3+\sqrt{10}}{6\sqrt{2\pi}}.$$

Theorem 1.1 (Sharp iid Berry–Esseen constant)**.** *For every integer $n \ge 1$,*

$$C_n \le C_{\mathrm{E}} \quad (n \ge 1), \qquad C^* = C_{\mathrm{E}} = \frac{3+\sqrt{10}}{6\sqrt{2\pi}}.$$

The upper bound follows from the ordinary analytical reductions proved here and in the precise source supplement [1] together with the verified exact numerical predicates of the thirteen families in Table 1; the matching lower bound $C^* \ge C_{\mathrm{E}}$ is given in Section 2.

The thirteen source families have been assembled, the scoped independent mathematical review is complete, and the full end-to-end public verification pass (`full_pass`) is established. The main argument below explains how their exact numerical conclusions enter ordinary mathematical implications. The companion `analytic-supplement.txt` contains the longer proofs, with original labels and a lemma-to-main-claim index. Verification results are summarized in Section 10.

Affine standardization needs no one-summand bound. For a law Z with mean μ, variance $\sigma^2 > 0$ and third central absolute moment β_3, set $X = (Z-\mu)/\sigma$. Then $\rho = \beta_3/\sigma^3$ and

$$\mathbb{P}\Big(\frac{\sum Z_i - n\mu}{\sigma\sqrt{n}} \le z\Big) = \mathbb{P}\Big(\sum X_i \le \sqrt{n}\,z\Big).$$

The result therefore has its usual arbitrary-mean/positive-variance form.

Throughout, H is a *raw sum* threshold. Its normalized version is $H/\sqrt{n}$. All terminating decimals used by a checker mean exact rationals. A numerical relaxation may contain nonrealizable moment points: soundness requires that it contain every actual law, not that every point be a law.

2 The independent classical lower approximation

Set $p = (4-\sqrt{10})/2$, $q = 1-p$, and $X = (B-p)/\sqrt{pq}$ for a Bernoulli variable B. Then $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$ and $\rho = (p^2+q^2)/\sqrt{pq}$. The fixed-law limit in Schulz [2, Remark 4.1, printed p. 15] gives, in the present normalization, $R_n(X) \to C_{\mathrm{E}}$: exact algebra gives $p^2+q^2 = 10-3\sqrt{10}$ and $(3+q-p)/(p^2+q^2) = 3+\sqrt{10}$. The threshold substitutions are

$$H = \sqrt{n}\,z, \qquad x = np + \sqrt{pq}\,H = np + \sqrt{npq}\,z.$$

Each ranges over the whole real line. For every $b < C_{\mathrm{E}}$, choose $\varepsilon = (C_{\mathrm{E}}-b)/2$ and a positive integer n with $R_n(X) > C_{\mathrm{E}} - \varepsilon > b$. Thus $C^* \geq C_{\mathrm{E}}$, independently of the upper proof and without finite-index attainment. The all-parameter finite-binomial upper theorem in [2, Theorem 1] is a separate input, used on the exact zero-noise branch below; neither theorem is proved by a PDF hash.

3 The unrestricted owner and its contact

Lemma 3.1 (Fixed-index attainment)**.** *For each fixed $n \geq 1$, $0 < C_n \leq \sqrt{n}$, and some law $P \in \mathcal{P}$ and threshold H attain C_n as a positive closed discrepancy:*

$$\mathbb{P}_P(S_n \leq H) - \Phi(H/\sqrt{n}) = \Delta_n(P), \qquad R_n(P) = C_n.$$

Proof. A negative closed discrepancy reflects into a positive strict discrepancy. Approaching that strict threshold from below, continuity of Φ gives the same supremum with positive closed discrepancies. Since $\rho \geq 1$, the upper bound is immediate. A symmetric two-point law, evaluated at the upper endpoint of its sum, gives positivity.

Take positive scores $r_j \to C_n$, with $r_j \geq c > 0$. Then $\rho_j \leq \sqrt{n}/c =: M$. Tightness gives a subsequence $P_j \Rightarrow P$; also take $\rho_j \to b$. The estimates

$$\int_{|x|>A} x^2\,dP_j \leq M/A, \qquad \int_{|x|>A} |x|\,dP_j \leq M/A^2$$

preserve the first two moments. Lower semicontinuity gives $1 \leq \rho(P) \leq b < \infty$. Chebyshev's inequality for negative thresholds, and the Gaussian upper tail for positive thresholds, imply that the chosen raw thresholds satisfy $|H_j| \leq n^{3/4}/\sqrt{c}$. Pass to $H_j \to H$. For fixed n, weak convergence of products and the continuous sum map give $P_j^{*n} \Rightarrow P^{*n}$. Closed-set Portmanteau at $H+\varepsilon$, followed by $\varepsilon \downarrow 0$, yields

$$F_{n,P}(H) - \Phi(H/\sqrt{n}) \geq C_n b/\sqrt{n}.$$

Consequently $C_n \geq R_n(P) \geq C_n b/\rho(P) \geq C_n$. All inequalities are equalities. In particular $\rho(P) = b$; third-moment convergence is a consequence of extremality, not an assumed uniform integrability of third moments. The proof includes $C_n = C_{\mathrm{E}}$. □

The law used below is the owner *selected by this lemma*, after orientation. We do not begin with an arbitrary law and pretend it remains an optimizer after extraction. Once selected, the law and raw threshold are fixed through every subsequent support, resource and count statement.

For $n \geq 2$, write $C = C_n$, $F = F_{n,P}(H)$, $z = H/\sqrt{n}$, $\gamma = \mathbb{E}X^3$, and $\beta = \mathbb{E}[X|X|]$. The exact contact statement used by the proof is

$$\rho I(y) = n^{3/2}\{F_{n-1,P}(H-y) - F\} + ny\phi(z) + \tfrac{1}{2}H\phi(z)(y^2-1) + \tfrac{3}{2}C\rho(y^2-1) - C\{|y|^3 - \rho - 3y\beta\}, \tag{1}$$

$$I(y) \leq 0 \quad (y \in \mathbb{R}), \qquad I(y) = 0 \quad (y \in \operatorname{supp} P). \tag{2}$$

Contact calculation. Contaminate the raw law by $P_\varepsilon = (1-\varepsilon)P + \varepsilon\delta_y$ and standardize. Its mean and variance derivatives are y and y^2-1. Its centered third-absolute-moment derivative is $|y|^3 - \rho - 3y\beta$, justified by an integrable quadratic derivative bound. The raw closed-convolution derivative is $n\{F_{n-1,P}(H-y) - F\}$, by the finite mixture expansion; the normal argument derivative is $-\sqrt{n}\,y - H(y^2-1)/(2\sqrt{n})$. Differentiating the normalized score gives exactly $\rho I(y)$; extremality makes it nonpositive. Its expectation is zero. The contact majorant is continuous while $F_{n-1,P}(H-y)$ is upper semicontinuous; thus strict negativity at a support point would give a negative integral on a neighborhood. This proves equality on the whole closed support. The cubic leading majorant confines that support before higher moments are introduced. See [1, lines410–463] for the full atom-safe variation and compactness details. □

4 The same-law predecessor step

For a standardized law P, define

$$q_n(P, C) = n \max\{R_{n-1}(P) - C, 0\}.$$

Lemma 4.1 (Weak predecessor is sufficient)**.** *If $C_{n-1} \leq C_{\mathrm{E}}$, then for every $P \in \mathcal{P}$ and every $C \geq C_{\mathrm{E}}$, $q_n(P, C) = 0$.*

Proof. The definition of the unrestricted supremum gives $R_{n-1}(P) \leq C_{n-1} \leq C_{\mathrm{E}} \leq C$. The positive part is zero. □

For the selected owner, use $C = R_n(P) = C_n$. Thus every previously proved implication using an allowance $q_n(P, C_n) \leq Q$, $Q \geq 0$, remains valid. No contact at $n-1$, strict predecessor, or monotonicity in n is inferred. Own-index score caps and resource bounds are separate premises and are not replaced by this lemma.

The source manuscript proves exactly this step at `indcore:lem:replacement`, lines 4777–4789. Some older resource descriptions assume the stronger $C_{n-1} < C_{\mathrm{E}}$. The assembly uses the actual weaker allowance hypothesis of their support/resource lemmas, together with Lemma 4.1; it does not assert that stronger predecessor.

5 Support and independently licensed Fourier bounds

Put $s = \sqrt{n/(n-1)}$, $e = n(s-1) - 1/2$, and $A = R_{n-1}(P)$. From contact one obtains, at a support point t,

$$0 \leq \mathcal{H}_n(H, t) + C\{-|t|^3 + \tfrac{3}{2}\rho t^2 + 3\beta t + \rho e\} + \rho s\, n(A - C).$$

Here

$$\mathcal{H}_n(H, t) = n\sqrt{n}\{\Phi((H-t)/\sqrt{n-1}) - \Phi(H/\sqrt{n})\} + \{nt + \tfrac{1}{2}H(t^2-1)\}\phi(H/\sqrt{n}).$$

For bounds $\rho \leq r$, $|\beta| \leq b$, $s \leq S$, $e \leq E$, $0 < c \leq C$, and $q_n(P, C) \leq Q$, define

$$Q_0(v) = -v^3 + \tfrac{3}{2}rv^2 + 3bv + rE, \qquad L(v) = \tfrac{2}{5}(v^2/2 + 1/2 + E) + cQ_0'(v).$$

The direct support lemma in [1], lines 492 onward, requires

$$Q_0(R), Q_0'(R), Q_0''(R), L(R), L'(R) < 0, \quad 1/5 - 3c < 0,$$

and, for both signs η,

$$\sup_{H \in \mathbb{R}} \mathcal{H}_n(H, \eta R) + cQ_0(R) + rSQ < 0.$$

Since $Q_0''' = -6$ and $L'' = 2/5 - 6c < 0$, these inequalities propagate along both complete closed rays $v \geq R$. They contradict contact on those rays, proving $\operatorname{supp} P \subset (-R, R)$. Whole-threshold Gaussian enclosures, including both infinite threshold tails, are required at the endpoint test. Sampling thresholds does not prove that supremum bound.

Fourier bounds serve different roles. An unrestricted auxiliary ceiling may license an own-score cap before support is known. A supported Fourier ceiling applies only after that same law's support has been acquired. Their normalization and endpoint terms must remain exact. In the 20–24 family the source Prawitz reduction uses the doubled kernel

$$K(t) = (1-t) + i\{(1-t)\cot(\pi t) + 1/\pi\}.$$

The four smoothing terms, the omitted endpoint pieces, and the infinite Gaussian tail are part of the certificate. A finite-window integral alone is not a Kolmogorov bound. The independent scalar-cosine certificate and Prawitz inequality are explicit inputs; numerical authentication cannot grant these analytic statements.

The exact iid literature inputs are [3]

$$R_n(P) \leq \min\{.4690,\ .3322(1 + .429/\rho),\ .3031(1 + .646/\rho)\}.$$

They apply for every $n \geq 1$: the source normalization for $X_i/\sqrt{n}$ has $\ell_n = \rho/\sqrt{n}$ and $\tau_n = 1/\sqrt{n}$. The n-generic compiled complement deliberately weakens .3322 to .3328; this also gives a valid bound, decreasing below .4097 at $\rho = 19/10$. The smoothing convention is fixed by [4, Lemma 4.1, printed p. 263]. The directed inequality $.4097 < C_{\mathrm{E}}$ is separate from the coarse $.4095 < C_{\mathrm{E}}$ used for cap/excess estimates.

6 One marginal and the conditional-count identity

Keep the same P. Set

$$p = \mathbb{P}(X > 0), \quad q = 1 - p, \quad v = pq, \quad m = \mathbb{E}X_+ = \mathbb{E}X_- > 0, \quad h = v/m.$$

The mean-zero, nondegenerate law has $0 < p < 1$. With $I = \mathbf{1}_{\{X>0\}}$, write

$$hX = I - p + U, \qquad \mathbb{E}(U \mid I) = 0.$$

A zero atom belongs to the nonpositive class. Let U_i be the two conditional laws and $w_i = \mathbb{E}U_i^2$. Then

$$s = qw_0 + pw_1 = h^2 - v, \qquad \lambda = ns.$$

In standardized physical units $\theta = s/(v+s)$, since $h^2 = v + s$. Consequently $\lambda = npq\theta/(1-\theta)$. If $s > 0$, let $\alpha = pw_1/s$; if $s = 0$, both conditional residuals are zero and no division by s is made.

Conditional on j positive draws, the sum of residuals has law

$$W_j = U_{0,1} + \cdots + U_{0,n-j} + U_{1,1} + \cdots + U_{1,j},$$

with independent summands. For $b_j = \binom{n}{j}p^j q^{n-j}$ and $np + hH = k + u$,

$$\begin{aligned}\mathbb{P}(S_n \le H) &= \sum_{j=0}^{n} b_j \mathbb{P}(W_j \le k + u - j) \\ &\le F_{\mathrm{Bin}(n,p)}(k) - b_k \mathbb{P}(W_k > u) + \sum_{j>k} b_j \mathbb{P}(W_j \le u - (j-k)). \end{aligned} \tag{3}$$

Indeed $hS_n = K - np + \sum U_i$; condition on K, retain $j = k$ exactly, and bound $j < k$ by its full mass. The outgoing event is strict because its complement is the closed event $\{W_k \le u\}$. Every incoming event remains closed. This identity works on both probability halves without reflecting p, β, γ.

Joint resources and units

Suppose the actual central support gap is $(-A, B)$. The scaled residual endpoints are

$$a_0 = p - hA, \quad a_1 = q - hB, \qquad U_0 \in [-O_0, a_0], \quad U_1 \in [-a_1, O_1].$$

The raw inward displacement is

$$d_{\mathrm{raw}} = (m/q - A) + (m/p - B), \qquad a_0 + a_1 = h\, d_{\mathrm{raw}}.$$

Thus raw inward displacement and its scaled COUNT coordinate cannot be identified without their factor h. In these scaled coordinates, a licensed joint plane has the form

$$\frac{a_0 + a_1}{d_0} + \frac{\lambda}{\Lambda} \le 1.$$

A later, smaller marginal cap for λ does not replace the intercept Λ in this plane. A zero resource budget has its own deterministic conclusion. The physical defect, its affine halfplane and the two signed quadratic constraints must all concern this same law. Their full proofs are the source labels `rescore:joint-resource`, `countcore:physicalD` and `countcore:Dhalfplane`; they are distinct ordinary analytical inputs, with proofs in that supplement.

7 What a finite COUNT certificate proves

On a complete acquired parameter box, the checker combines valid outgoing lower and incoming upper probability bounds in (3), then subtracts a lower Gaussian bound at the *same* phase. If B_k, b_k^-, b_j^+, L_k, A_j, N denote those bounds, it certifies a whole-box upper for

$$B_k - b_k^- L_k + \sum_{j>k} b_j^+ A_j - N.$$

This combined expression retains phase cancellation. Separate maxima at unrelated phases would not be the same proof. The positive error is then multiplied by an upper enclosure of $\sqrt{n}$ and divided by a positive lower enclosure of the same ρ. Every closed normalization piece is included. Convex vertex maximization is performed before minimizing over a finite menu of strategies; a pointwise minimum is not assumed convex.

If the acquired bounds imply

$$np_- - h_+ H_- \ge -1/2, \qquad np_+ + h_+ H_+ \le n + 1/2,$$

every threshold has a nearest count $k = 0, \dots, n$ and $u \in [-1/2, 1/2]$. Otherwise the remaining thresholds require separate proved tail bounds. The count notation does not discard them.

Each of the $n+1$ original product roots is covered by a finite binary tree. A strict interior rational cut retains *both closed children*. Every leaf either proves a necessary-condition contradiction or a whole-box normalized score upper below a rational $c < C_{\mathrm{E}}$. Induction up the finite tree proves its root statement; union over all roots proves the complete comparison, including ties, boundaries and zero-noise faces. Missing roots, unsuccessful guards and unfinished searches are not exclusion certificates. This is the precise finite closed-cover implication of [1], `countcore:scoretheorem` and `countcore:forest`.

The entire matching contract and physical resource-to-COUNT implication remain mandatory. A local COUNT acceptance does not acquire support, an own-index cap, contact, or a new all-law theorem by itself.

8 The literal 20–24 chain

This completed source block makes the assembly concrete. The full numerical role set has 186 entries: 163 ordinary forests, eight INITIAL/resource cores, three Fourier packets, one same-law 19/20 feedback packet, one high-support packet, the shared scalar certificate, one full actual20 conjunction and eight full band joins. Already checked inputs are reused with their existing provenance; this list is not a request to repeat their numerical work.

For each $n = 21, \dots, 24$, the closed low-rho bands and initial support radii are

	ρ interval	radius
$B0$	$[1, 11/10]$	$23/10$
BN	$[11/10, 9/8]$	$12/5.$

Every own cap 419/1000 comes from a separate unrestricted auxiliary Fourier bound at the *actual* index. Each core starts without an acquired defect seed, runs eight feedback covers and a ninth export, and accounts for all 2592 signed cells per cover and all 100 closed probability panels. Its quantitative resource meaning is required even if a discovery target happened to be reached earlier.

Actual20 uses 49 complete forests, with its original whole-panel two/four-piece unions. The eight later bands use 114 forests and 800 complete probability dispositions. All 1000 panels in this five-index block are accounted for by physical contradiction, strict scalar contradiction, or the full matching COUNT. Every probability seam is closed. The high-rho complement uses supported Fourier on $[9/8, 6/5]$, support-free Fourier on $[6/5, 19/10]$, and the full published infinite tail. Radius-3 support for that supported branch is a separate premise.

Proposition 8.1 (Conditional five-index composition)**.** *Suppose the actual all-law $C_{19} \le C_{\mathrm{E}}$ theorem is supplied, every numerical input above passes its complete mathematical interpreter, and the exact attainment/contact, Fourier, support, resource, realization and cover implications are proved or accepted at their stated scope. Then $C_n \le C_{\mathrm{E}}$ for $n = 20, \dots, 24$.*

Proof. At the next literal index assume $C_n > C_{\mathrm{E}}$, choose the unrestricted owner by Lemma 3.1, and freeze that law and raw threshold. Lemma 4.1 licenses the same-law predecessor allowance. The separate actual-index own caps, full rays, INITIAL and resource implications place the owner in the complete low/high cover. Every part contradicts its score or a necessary physical condition. Thus $C_n \le C_{\mathrm{E}}$. Repeat at the next integer. No desired next bound, sample-size monotonicity, or contact at a transferred index is an input. □

The verified release supplies the actual whole41 predicate, all 49 `actual20` COUNT jobs and their mathematical namespace bridge, the 114 public COUNT forests for indices 21–24, the completed 185 resource/scalar meanings, the genuine incoming19 theorem and all 12 analytical implications.

9 The thirteen roles and the unbounded tail

Table 1: Exact upper partition. Every finite index1–48 occurs once.

Index	Mathematical role	Complete source/numerical meaning
1	$C_1 \le 2/5 < C_{\mathrm{E}}$; Appendix A.	150 closed cubic-dual cells, independent exact checks.
2	$C_2 \le 2\sqrt{2}/7 < C_{\mathrm{E}}$, full unrestricted tensor reduction.	Two independent complete polynomial checkers.
3	$C_3 \le 59\sqrt{3}/250 < C_{\mathrm{E}}$, 59-feature null polynomial.	Full scalar, identity and cube predicates.
4	$C_4 \le .4097 < C_{\mathrm{E}}$ from quantitative G3.	Both ROOT4 whole260-parent passes, HIGH/LOW and reserves.
5	New one-free G3→G4→same-owner $q_5 = 0$ contradiction.	Whole region/resource/COUNT cover;13 retained rules, no .3704 premise.
6–9	Exact source implication with the new proof of the identical weak C5 parent.	Whole original contexts, explicit parent substitution.
10–19	Exact parent substitution from the genuine6–9 theorem.	Whole original resource/COUNT contexts and closed subdomains.
20–24	Increasing weak induction, actual own caps < .419.	186 meanings,12 analytical rules,163 COUNTs,1000 panels.
25	New low-only $[1, 6/5]$ cover and independent unrestricted high-rho branch.	Six original calls,52 complete COUNT interfaces, zero noise.
26–39	New low-only weak induction; actual support35 and separate positive39 cap.	300 COUNTs, whole7 partitions,35 support predicates, physical interfaces.
40–42	New floor40 theorem for actual $n \in \{40, 41, 42\}$.	CORE304+2/9/INITIAL2; honest original8+new2 acquisition conjunction.
43–48	Independent positive ceilings and finite least failure, no sharp42 premise.	Original183 complete parents; distinct floor42 LOW family.
$n \ge 49$	Independent positive ceilings, strong46, uniform local exclusion.	14 top-level plus41 retained execution predicates; full closed cover and rays.

The principal finite dependency is G3→G4→C5→C6–9→C10–19 →C20–24→C25→C26–39→C40–42. N1 and G2 are independent. The unrestricted high-rho25–48 theorem is independent. The43–48 and49-tail roles have their own positive/coarse prerequisites: no sharp42→43 edge and no sharp1–48 premise in the tail are used. Proof-parent substitution requires an identical statement and retains the complete numerical context.

Weighted floors and signed corrections

A floor row N is applied only with $nD_{\mathrm{actual}} \le Nd$ and both the defect ceiling and noise price scaled by N/n; see Appendix B. All signed moment, support, probability and threshold fields must still match. For real I and $40 \le n \le 42$, $I/n \le \max(I/40, I/42)$. For a signed bracket B

and multiplier in $[a, b]$, its lower product is $\min(aB, bB)$. These are affine endpoint inequalities, requiring no positive sign for I or B. The paired portfolio retains $M < 3$, $g^2 < 2$, $R > 2g$ and its positive damping gates. Negative skewness uses the upper density endpoint, or its explicit equivalent correction, before completing squares. Both raw and barrier-subtracted curvatures are corrected. The ordinary active-override proof, licenses this reuse.

Three losses at49 and the least later failure

The independent auxiliary branch gives

$$C_j \le C_{\mathrm{E}} + \frac{.7}{j} \ (45 \le j \le 50), \qquad C_{46} < C_{\mathrm{E}} + \frac{.54}{46}.$$

Write $d_n = n(C_{n-1} - C_n)_+$. Its uniform local theorem excludes a positive contrary owner for every actual $n \ge 47$ with excess at most $.7/n$ and $d_n \le .19$, current cap $C_n < .4247$, and deleted caps $C_{n-1}, C_{n-2} < .4253$. The entry theorem acquires the same-law support, probability, moment and resource hypotheses independently of sharpness. Conditional-count, signed-deletion, confinement, marked-crossing and scaled-small-noise arguments prove $\rho_B(C - C_B) < -\lambda/1000$ for every $\lambda > 0$. At $\lambda = 0$, centering gives $U = 0$ almost surely and Schulz's separate finite-binomial theorem gives $C_B < C_{\mathrm{E}}$. The full ordinary proofs are [1, lines7647–8064 and8217–8887]; their complete numerical predicates, signed gates and closed seams are required, not inferred from a PASS or finite sampling.

If $C_{49} > C_{\mathrm{E}}$, the rational caps license the local theorem successively at49,48,47. Each failure forces $C_{n-1} > C_n + .19/n$. Hence

$$C_{46} > C_{\mathrm{E}} + \frac{19}{100}\left(\frac{1}{47} + \frac{1}{48} + \frac{1}{49}\right) > C_{\mathrm{E}} + \frac{27}{50 \cdot 46},$$

since the excess difference is exactly $35419/254251200 > 0$. This proves weak sharpness at49 without sharp1–48. Each application may choose its own owner; its internal same-law premises never change.

For the least later violation $n \ge 50$, the previous weak sharp theorem gives zero same-law decrement. The support-free smoothing bound is

$$C_n \le \sqrt{\frac{n}{n-1}} C_{n-1} + \frac{\sqrt{n}}{15(n-1)^{3/2}} + \frac{2}{25}\frac{\sqrt{n}}{(n-1)^2}.$$

It follows by conditioning on one summand and using third-absolute-moment Gaussian Taylor and variance heat remainders before either supremum [1, lines7463–7506]; no support or higher moment is assumed. For $n \ge 50$, $C_{\mathrm{E}} < 409733/10^6$ and $\sqrt{n/(n-1)} < 1011/1000$ give

$$n(C_n - C_{\mathrm{E}}) < \frac{409733}{10^6}\frac{25}{49} + \frac{50}{49}\frac{1011}{15000} + \frac{2}{25}\frac{50}{49}\frac{1011}{7000} < .29.$$

Thus the local theorem applies again. At $n = 50$ its second predecessor48 uses the independent coarse bound $C_{48} \le C_{\mathrm{E}} + .7/48 < .4253$; thereafter both deleted indices are earlier sharp ones. Contradiction proves the entire unbounded tail. The thirteen-part cover and the separate lower approximation prove Theorem 1.1 under its stated predicates.

10 Verification status

The original source computations, global mathematical composition, scoped independent analytical/checker-source review, and full end-to-end public verification run (`full_pass: "ESTABLISHED"`) are complete. All numerical hypotheses in Theorem 1.1 are verified by complete source-bound executions across all thirteen mathematical roles: the 150-cell exact

cubic-dual certificate at $n = 1$ (`N1`); the two independent polynomial checkers at $n = 2$ (`G2`), the 59-feature polynomial and 111-cell C++ cube checker at $n = 3$ (`G3`), and the complete two-pass 260-geometry verification at $n = 4$ (`G4/ROOT4`); all 1,243 public `COUNT` jobs and all 49 `actual20 COUNT` jobs with their namespace bridge and 5 scalar contracts; all 183 native `finite43-48` tasks and the two collectors (`collector43`, `collector44`); and all 55 native `tail49` predicates (14 top-level plus 41 retained execution predicates). Running `python -I -B be.py status` and `python -I -B be.py verify` verifies the complete release with zero remaining obligations (`EXIT_CODE=0`).

The accompanying ordinary proofs establish the analytical implications; execution receipts in `verification-results/` identify the verified numerical computations. Complete Lean formalization of the analysis and checker semantics remains future work; the existing assembly/limit bridges verify the global mathematical composition. AI assistance was used in development, source review and preparation of this manuscript.

A The one-summand dual proof

For each rational cell $[a, b]$ of a closed cover of $[-2, 1]$, the finite input supplies $u, v, L_a \in \mathbb{Q}$, $L_a \le \Phi(a)$, with

$$L_a - v + ux + vx^2 + \tfrac{2}{5}|x|^3 - \mathbf{1}_{\{x\le b\}} \ge 0 \quad (x \in \mathbb{R}).$$

Whole-line polynomial/Sturm checks include both infinite rays. Expectation cancels $-v+v\mathbb{E}X^2$ and $u\mathbb{E}X$, giving $F_X(b)-L_a \le (2/5)\rho$. Monotonicity gives the same upper bound for $F_X(t)-\Phi(t)$ on $[a, b]$. At $t \le -2$, Cantelli gives $F_X(t) \le 1/5$; at $t \ge 1$, $1 - \Phi(1) < 2/5$ (for example $\phi(0) > 3/8$ and $e^{-s^2/2} \ge 1 - s^2/2$ give $\Phi(1) > 13/16$). Since $\rho \ge 1$, these suffice. Finally $\Phi(t) - F_X(t) \le F_{-X}(-t) - \Phi(-t)$, since $F_{-X}(-t) = 1 - F_X(t-)$, so reflection includes atoms. The150-cell certificate and independent exact checks are finite inputs; this ordinary implication needs no general-law .3704 claim.

B Physical defect and weighted units

Put $\sigma = \sqrt{10}$, $\tau(t) = \mathbb{E}[X\mathbf{1}_{\{X\ge t\}}]$ for the bounded owner. Then $\tau \ge 0$, $\int \tau = 1$ and $2\int |t|\tau(t)dt = \rho$. For complementary gaps (a, b) of its closed support let $\tau_{a,b}$ be the constant interior value, and put

$$J = \sum_{(a,b)} \tau_{a,b}(b-a)^2, \qquad D = (3+\sigma)\rho - \gamma - 3J.$$

The sums converge by Tonelli and bounded support. If zero is in the support, $(b-a)^2 \le 2\int_a^b |t|dt$ gives $J \le \rho$ and $D \ge 0$. Otherwise let $(-A, B)$ be the central gap, $m = \mathbb{E}X_+ = \mathbb{E}X_-$, $T_+ = \mathbb{E}X_+^3 - mB^2$, $T_- = \mathbb{E}X_-^3 - mA^2$, $J_o = J - m(A+B)^2$. On positive outer gaps use $(b-a)^2 \le 2\int_a^b (t-B)dt$ and reflect; thus $J_o \le T_+ + T_-$. Direct expansion gives

$$D = m(\sqrt{\sigma+1}A - \sqrt{\sigma-1}B)^2 + (\sigma-1)T_+ + (\sigma+1)T_- + 3(T_+ + T_- - J_o) \ge 0.$$

For the actual conditional means $\bar{A} = m/q$, $\bar{B} = m/p$, write $\Delta_A = \bar{A} - A$, $\Delta_B = \bar{B} - B$ and let θ_+, θ_- be the positive/negative conditional centered second moments with class weights included; $\theta = \theta_+ + \theta_-$. Conditional cubic expansion and the stronger outer estimates give the simultaneous ordinary budget

$$\begin{aligned} D \ge{}& \frac{(6+2\sigma)(1-\theta)^{3/2}}{\sqrt{pq}}(p-p_*)^2 + 6m\{(A+B)(\Delta_A+\Delta_B) + \Delta_A\Delta_B\} \\ &+ 3(\sigma+1)B\theta_+ + 3(\sigma+3)A\theta_-, \qquad p_* = (4-\sigma)/2. \end{aligned}$$

Indeed the positive cubic residual is at least $3B\theta_+$ and its reflection handles the other sign. The deterministic remainder is exactly the second term, and $(\sigma+1)p^2+(\sigma-1)q^2-6pq = (6+2\sigma)(p-p_*)^2$ with $m^2=(1-\theta)pq$ proves the binary term. Stronger signed/mean/outer-gap refinements retain their full ordinary proofs in [1, lines2470–4039]; $D\ge 0$ alone would not establish them.

If $n\ge N>0$, all costs are nonnegative and

$$(n/N)D_{\text{actual}}\le d, \qquad D_{\text{actual}}\ge b+gd_s+(N/n)c_\lambda\lambda,$$

then $d\ge(n/N)(b+gd_s)+c_\lambda\lambda\ge b+gd_s+c_\lambda\lambda$. This proves the basic weighted transfer. A later smaller marginal noise ceiling does not replace the original joint intercept. Negative residuals exclude; a zero residual forces nonnegative coordinates to zero.

C Normalization across one half

The original sufficient binomial recipe required a Taylor segment below 1/2. The active recipe permits crossing under the following exact extension, independently checked in the final audit.

Lemma C.1 (Symmetric derivative envelopes)**.** *If $0<a<1/2$, $a<b<1$, $a+b<1$, then for $r\in[a,b]$,*

$$r(1-r)\ge a(1-a), \qquad \frac{|1-2r|}{[r(1-r)]^j}\le\frac{1-2a}{[a(1-a)]^j}\quad (j=1,2).$$

Proof. $[a,b]\subset[a,1-a]$. Set $s=\min(r,1-r)\in[a,1/2]$. The expressions are symmetric; $s(1-s)$ increases and $1-2s$ decreases nonnegatively. Their positive quotients therefore have their maxima at a. □

Let $n\ge N\ge 1$, $p\in[p_-,p_+]\subset(0,1/2)$, $q=1-p$, $1\le k\le n-1$, $\delta=k-np$, $r=k/n$, $v=pq$. Suppose the whole segment between p,r lies in $[a,b]$ above. If

$$E\ge\frac{1-2a}{a^2(1-a)^2},\quad L\ge\frac{1-2a}{a(1-a)},\quad R\ge\frac{1}{12a(1-a)},$$

then the binomial mass π_k satisfies

$$\sqrt{n}\,\pi_k=\frac{\phi(0)}{\sqrt{v}}e^{-\delta^2/(2nv)+\zeta}, \qquad n|\zeta|<L|\delta|/2+R+E|\delta|^3/(6N).$$

For proof, relative entropy has $D(p\|p)=D'(p\|p)=0$, $D''(p\|p)=1/v$ and $D^{(3)}(r\|p)=(2r-1)/[r^2(1-r)^2]$. Taylor bounds $|nD(r\|p)-\delta^2/(2nv)|$ by $E|\delta|^3/(6n^2)$; the logarithmic variance prefactor changes by at most $L|\delta|/n$. Finite Stirling, $0<s_j<1/(12j)$, gives exactly

$$\sqrt{n}\,\pi_k=\frac{\phi(0)}{\sqrt{r(1-r)}}e^{-nD(r\|p)+s_n-s_k-s_{n-k}}.$$

Writing $e=s_n-s_k-s_{n-k}$, use the compatible one-sided estimates $e<1/(12n)$ and $e>-1/(12k)-1/(12(n-k))=-1/[12nr(1-r)]$. Thus $|e|<R/n$, and collecting all three errors proves the claim. One must not add three incompatible absolute factorial estimates.

The literal code gates take $a=$ `left`, $b=$ `right`, $a<p_--\Delta/N$, $p_++\Delta/N<b$, $a+b<1$. For $|\delta|\le\Delta$ they contain the whole segment. Gates $Np_--\Delta-2>0$ and $N(1-p_+)-\Delta-2>0$ ensure interior/adjacent factorial domains. Positive $v_f<a(1-a)$ and upward-rounded $(1-2a)/v_f^2$, $(1-2a)/v_f$, $1/(12v_f)$ supply E,L,R. Mass bounds use $e^{-x}\ge 1-x$, $e^x\le 1/(1-x)$ only on their stated positive domains. The original p band remains below 1/2 when bounding pq; monotonicity is not assumed on the enlarged Taylor segment. This lemma grants no physical acquisition or numerical exclusion.

D Exact source and audit map

The ordinary source supplement [1] has SHA-256

`2b5c9419a5fc2363de21e02e200529b735`
`c52e492f0d398aca6f42650c65eea1.`

Exact applicable sections are attainment/contact368–463; Fourier/resource1031–4039; physical COUNT4043–4765; small-index4822–6623; and smoothing7458–7506 plus confinement/local7647–8887. These are ordinary theorem/proof references, not numerical acceptance authorities. The longer calculations are provided in `analytic-supplement.txt`, whose opening index maps their lemmas to the claims used in this paper.

The completed review is `work/final-audit/INDEPENDENT-MATHEMATICAL-AUDIT-v1.md`, incorporating arithmetic continuations v1–v5, the resource-refinement review and the native HIGH/support/barrier/confinement review. The actual global upper/lower/equality objects have SHA prefixes `5acc019642ae`, `018bdf1b6918`, `8be2f7c6083d`; the full source/witness manifest supplies their identities. Hashes locate objects; the arguments and complete predicates give meaning. The companion supplement gives the ordinary proofs and their exact source scopes.

References

[1] K. Rodziewicz and B. Kołodziejek, *The iid Berry–Esseen constant: Analytic reductions and finite verification*, working analytical source manuscript, October2026. Selected ordinary proof sections are reproduced in the accompanying `analytic-supplement.txt`, with their original labels and a dependency index.

[2] J. Schulz, *The optimal Berry–Esseen constant in the binomial case*, Dissertation, Universität Trier,2016. Theorem1 and Remark4.1, printed p. 15 (PDF p. 25).

[3] I. Shevtsova, *On the absolute constants in the Berry–Esseen inequality and its structural and nonuniform improvements*, Informatics and Applications7(1) (2013),124–125 (in Russian).

[4] I. Shevtsova, *Moment-type estimates with asymptotically optimal structure for the accuracy of the normal approximation*, Annales Mathematicae et Informaticae39 (2012),241–307; Lemma4.1, printed p. 263.